Shobhita Sundaram

shobhita@mit.edu
ssundaram21.github.io

Education

Massachusetts Institute of Technology (MIT) — Cambridge, MA
Ph.D. Computer Science — 2022–Present
S.M. Computer Science — 2022–2025
Advisor: Phillip Isola
Research: Self-improving systems; scaling capabilities with synthetic data.

Massachusetts Institute of Technology (MIT) — Cambridge, MA
S.B. Computer Science, S.B. Mathematics — 2018–2022
Advisors: Pawan Sinha, Xavier Boix, Tomaso Poggio

Experience

Meta Superintelligence Labs — Paris, France
Research Scientist Intern — May - October 2025

- *Project:* Asymmetric self-play with LLMs for difficult reasoning problems.
- *Mentors:* Julia Kempe, Yann Ollivier, Kartik Ahuja

Google Research — Cambridge, MA
Student Researcher — December 2023 - March 2024

- *Project:* Synthetic data generation with diffusion models for personalizing vision backbones.
- *Mentors:* Yonglong Tian, Dilip Krishnan

Google DeepMind — London, UK
Research Engineering Intern — June - August 2022

- *Project:* Data selection for LLM pretraining.
- *Mentors:* Sebastian Borgeaud, Laurent Sifre, Jordan Hoffmann, Arthur Mensch

Center for Brains, Minds, and Machines, MIT — Cambridge, MA
Undergraduate Researcher — September 2019 - May 2022

- *Project:* Recurrent vision models for visual long-range spatial dependencies.
- *Mentors:* Xavier Boix, Pawan Sinha, Tomaso Poggio

The D. E. Shaw Group — New York, NY
Quantitative Research Intern — June - August 2021

- *Project:* RL for portfolio management; outperformed optimal control approaches.
- *Mentor:* Konstantin Turitsyn

Apple — Cupertino, CA
Machine Learning Intern — June - August 2020

Two Sigma Investments — Houston, TX
Software Engineering Intern — May - August 2019

Publications

** indicates equal contribution*

1. Do LLM Agents Still Need to Be Told How to Self-Improve?
   S. Sundaram, C. Sancaktar, P. Isola. *Under review*, 2026.

2. Teaching Models to Teach Themselves: Reasoning at the Edge of Learnability.
   S. Sundaram, J. Quan, A. Kwiatkowski, K. Ahuja, Y. Ollivier, J. Kempe. *ICML*, 2026 **(spotlight)**.

3. Better Together: Leveraging Unpaired Multimodal Data for Stronger Unimodal Models.
   S. Gupta, **S. Sundaram**, C. Wang, S. Jegelka, P. Isola. *ICLR*, 2026.

4. What Makes for a Good Stereoscopic Image?
   N. Tamir, S. Amir, R. Itzhaky, N. Atia, **S. Sundaram**, S. Fu, R. Sokolovsky, P. Isola, T. Dekel, R. Zhang, M. Farber. *CVPR Computer Vision for Metaverse Workshop*, 2025.

5. Personalized Representation from Personalized Generation.
   S. Sundaram*, J. Chae*, Y. Tian, S. Beery, P. Isola. *ICLR*, 2025.

6. When Does Perceptual Alignment Benefit Vision Representations?
   S. Sundaram*, S. Fu*, L. Muttenthaler, N. Tamir, L. Chai, S. Kornblith, T. Darrell, P. Isola. *NeurIPS*, 2024.

7. DreamSim: Learning New Dimensions of Human Visual Similarity using Synthetic Data.
   S. Fu*, N. Tamir*, **S. Sundaram***, L. Chai, R. Zhang, T. Dekel, P. Isola. *NeurIPS*, 2023 **(spotlight)**.

8. Recurrent Connections Facilitate Symmetry Perception in Deep Networks.
   S. Sundaram*, D. Sinha*, M. Groth, T. Sasaki, X. Boix. *Scientific Reports*, 2022.

9. GAN-Based Data Augmentation for Chest X-ray Classification.
   S. Sundaram* and N. Hulkund*. *KDD Workshop on Applied Data Science for Healthcare*, 2021.

10. Do Neural Networks for Segmentation Understand Insideness?
    K. Villalobos*, V. Štih*, A. Ahmadinejad*, **S. Sundaram**, J. Dozier, A. Francl, F. Azevedo, T. Sasaki, X. Boix. *Neural Computation*, 2021.

Awards

NSF Graduate Research Fellowship	2022 - 2025
HDTV Grand Alliance Fellowship	2022 - 2023
MIT Undergraduate Research and Innovation Scholar	2020
MIT Burchard Scholar	2020

Service & Leadership

Organizer: CVPR Workshop on Synthetic Data for Computer Vision	2024 - 2026
Organizer: ECCV Tutorial on Efficient Text-to-Image Modeling	2024
Mentor: MIT Graduate Application Assistant Program	2022 - 2024
Event Coordinator: MIT Graduate Women of EECS	2023
Associate Editor: MIT Science Policy Review	2020 - 2022

Invited Talks

Self-Improving at the Frontier of Learnability.
Lila Sciences — April 2026
The Vector Institute — February 2026

Personalized Representation from Personalized Generation.
CVPR Syntagen Workshop — June 2025
Cohere for AI — March 2025

Representation Learning with Perceptual Alignment.
Stanford NeuroAILab — April 2025

Evaluating Text-to-Image Models.
ECCV Efficient Text-to-Image Modeling Tutorial — September 2024

DreamSim: Learning New Dimensions of Human Visual Similarity using Synthetic Data.
Adobe — October 2023
Computer Vision Meetup, hosted by Voxel51 — July 2023